

TORAL BANERJEE

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EDUCATION

Clark University, Worcester, MA

B.A. Computer Science, Mathematics

Fall 2024 — Expected May 2028

GPA: 3.64 / 4.0 • **Atupem Award (\$2,000)** — granted annually to one student showing exceptional motivation toward an impactful summer internship.

EXPERIENCE

HCI Research Assistant — AI Companion Safety, AI4UGC Lab, Clark University

Jun 2026 – Present

- Building a structured coding scheme and labeling workflow to categorize user disclosures, distress signals, and parasocial-attachment patterns across mental-health-adjacent online communities, supporting an HCI research submission on companion-AI safety harms.
- Designing a harm taxonomy and resolving boundary cases (active vs. retrospective episodes, first- vs. third-person framing) to ensure reproducible, inter-rater-aligned annotation across a multi-thousand-post corpus.
- Translating qualitative HCI insight into machine-readable categories that feed downstream LLM-based evaluation of companion-AI guardrails, bridging user-facing harm patterns with applied AI safety auditing.

HCI Research Assistant — SocraTalk, Prof. Hyungsin Kim Lab, Clark University

Jan 2026 – Present

- Building a browser-instrumentation pipeline that captures DOM snapshots and user-interaction events, generating high-signal behavioral traces for retrieval-based evaluation and analysis of infinite-scroll patterns among teenage users.
- Implementing agent-driven flows with prompt orchestration and tool-calling logic to enable autonomous system interactions during user experimentation.
- Managing lightweight in-context memory to preserve session state across user studies.

Undergraduate Research Assistant — AISL Lab

Jan 2025 – Jan 2026

- Authored first-author correspondence published in *Nature* on AI regulation and consent in medical AI; preparing follow-up commentary for *Science* on AI safety in genomics.
- Developed ML surrogate models and agent-assisted workflows to replace expensive physics simulations, accelerating experiment-cycle and iteration speed.
- Designed multi-physics workflows (polarization, band engineering, carrier transport) improving 2DEG density by 5.8%.

Data Analytics Intern — Reliance BP Mobility Ltd., Mumbai, India

Summer 2025

- Built a customer-segmentation ML pipeline across 1,400 fuel outlets, applying K-Means and RFM clustering on 15,000+ customers.
- Designed Tableau and Matplotlib dashboards translating raw transactional data into stakeholder-facing insights.
- Drove a 10% reduction in customer churn and a 15% gain in redemption efficiency through targeted segmentation.

Teaching & Peer Learning Assistant — Data Structures, Pre-Calculus, Clark University

Jan 2026 – May 2026

- Led problem-solving, debugging, and office-hours support across two courses; partnered with faculty to identify recurring conceptual gaps and design targeted intervention exercises.

PROJECTS

Agent Trajectory Inspector — agent-eval visualizer for web-navigating agents

2026

React · Vite · TypeScript · Groq (Llama 3.3 70B) · deployed on Vercel

- Built because agent failures stay opaque and post-hoc — developers see the outcome of an agent run but not the decision path that produced it, so failures get patched without ever being understood.
- Surfaces step-level reasoning, tool calls, and manipulation-tactic annotations across the full trajectory, making opaque agent runs inspectable and debuggable.
- Designed for agent-evaluation workflows at safety-eval startups like Patronus AI. Live at agent-trajectory-inspector.vercel.app.

Decision Receipt — transparency layer for AI-driven loan decisions

2026

Next.js · TypeScript · Tailwind · Groq · counterfactual generation, proxy-discrimination detection

- Built for the person on the receiving end of an AI loan denial — a one-line rejection leaves users no way to verify, appeal, or contest the decision, while the lender sees the full reasoning.
- Turns opaque ML outputs into inspectable, contestable artifacts: counterfactual generation, appeal-letter drafting, proxy-discrimination detection, and peer comparison.
- Positioned against an emerging regulatory landscape (EU AI Act, CFPB); competitive scan surfaced no direct consumer-facing alternative.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, R

AI & Agents: LLM orchestration (OpenAI, Anthropic, Groq), tool-calling logic, prompt engineering, agent-driven workflows, agent evaluation, RAG

ML & Data: PyTorch, Transformers (HuggingFace), NumPy, predictive modeling, Apache Spark, Tableau, Matplotlib

Frontend & Infra: Next.js, React, Vite, Tailwind, Vercel, Git, Linux/Bash, PostgreSQL, MySQL